The Exponent Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: RULES, NEGZERO, SCINOT and EXPFUNC (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (RULES)
- ☐ **A2** (NEGZERO)
- ☐ **A3** (SCINOT)
- ☐ **A4** (EXPFUNC)
- ☐ **A5** (RULES)
- ☐ **A6** (SCINOT)

Entrées

10 pts each · choose 4

- ☐ **E1** (RULES)
- ☐ **E2** (NEGZERO)
- ☐ **E3** (SCINOT)
- ☐ **E4** (EXPFUNC)
- ☐ **E5** (NEGZERO)
- ☐ **E6** (EXPFUNC)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (RULES)
- ☐ **S2** (SCINOT)
- ☐ **S3** (EXPFUNC)
- ☐ **S4** (NEGZERO)

My skill check (circle each skill you have covered): RULES · NEGZERO · SCINOT · EXPFUNC

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 RULES ☐ My pick · 5 pts

Simplify: $(3x^2)(4x^5)$

A2 NEGZERO ☐ My pick · 5 pts

Evaluate each: **(a)** 2^{-3} **(b)** $6^0 + 3^{-2}$

A3 SCINOT ☐ My pick · 5 pts

Write in scientific notation: **(a)** 52,000 **(b)** 0.0006

A4 EXPFUNC ☐ My pick · 5 pts

Complete the table for $y = 3 \cdot 2^x$ for x = 0, 1, 2, 3. Is this growth or decay?

A5 RULES ☐ My pick · 5 pts

Simplify: **(a)** $\frac{x^8}{x^3}$ **(b)** $(y^3)^4$

A6 SCINOT ☐ My pick · 5 pts

Which is larger, 3.2×10^5 or 9.1×10^4? Explain how you know.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 RULES ☐ My pick · 10 pts

Simplify, writing with positive exponents:

(a) $\dfrac{12x^5y^2}{18x^2y^6}$

(b) $(2a^3b^{-2})^3$

E2 NEGZERO ☐ My pick · 10 pts

Simplify with positive exponents:

(a) $\dfrac{5x^{-3}}{y^{-2}}$ **(b)** $(3x^{-2})^2$ **(c)** $4m^0 + (4m)^0$

E3 SCINOT ☐ My pick · 10 pts

Compute and write each answer in scientific notation:

(a) $(3 \times 10^4)(5 \times 10^7)$

(b) $\dfrac{8.4 \times 10^5}{2.1 \times 10^{-2}}$

E4 EXPFUNC ☐ My pick · 10 pts

For x = 0, 1, 2, 3 a table shows y = 80, 120, 180, 270. **(a)** Show the table is exponential. **(b)** Write an equation $y = a \cdot b^x$. **(c)** Find y when x = 4.

E5 NEGZERO ☐ My pick · 10 pts

Kai wrote:

(a) $3^{-2} = -9$ **(b)** $6x^0 = 1$

Each answer has a mistake. Explain each mistake and give the correct value.

E6 EXPFUNC ☐ My pick · 10 pts

A $10,000 car loses 20% of its value each year. **(a)** Write an equation for its value V after t years. **(b)** Find its value after 3 years. **(c)** Has it lost more than half its value after 3 years? Justify.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 RULES ☐ My pick · 20 pts

(a) Simplify completely, with positive exponents:

$$\frac{(3x^2y)^3 \cdot 2x^{-1}y^4}{6x^4y^{-2}}$$

(b) Evaluate your answer when $x = 2$ and $y = 1$.

(c) Name the exponent rules you used.

S2 SCINOT ☐ My pick · 20 pts

Light travels 3×10^8 meters per second. The Sun is 1.5×10^{11} meters from Earth. **(a)** How many seconds does sunlight take to reach Earth? **(b)** A spacecraft moves 3×10^4 meters per second. How long would it take to reach the Sun’s distance? **(c)** How many times longer is (b) than (a)? Use scientific notation.

S3 EXPFUNC ☐ My pick · 20 pts

A bacteria culture starts with 500 cells and doubles every 3 hours. Let h be the number of 3-hour periods. **(a)** Find the cells after 12 hours. **(b)** What is the first time (in hours) the count exceeds 100,000? **(c)** Another culture has 12,000 cells now and doubles every 3 hours. How many cells were there 9 hours ago?

S4 NEGZERO ☐ My pick · 20 pts

(a) The pattern $2^4 = 16$, $2^3 = 8$, $2^2 = 4$, $2^1 = 2$ continues. Find 2^0, 2^{-1}, 2^{-2} and explain why each makes sense.

(b) Simplify with positive exponents: $\dfrac{x^{-3}y^2}{x^{-5}y^{-1}}$

(c) Solve: $3^x = \dfrac{1}{81}$